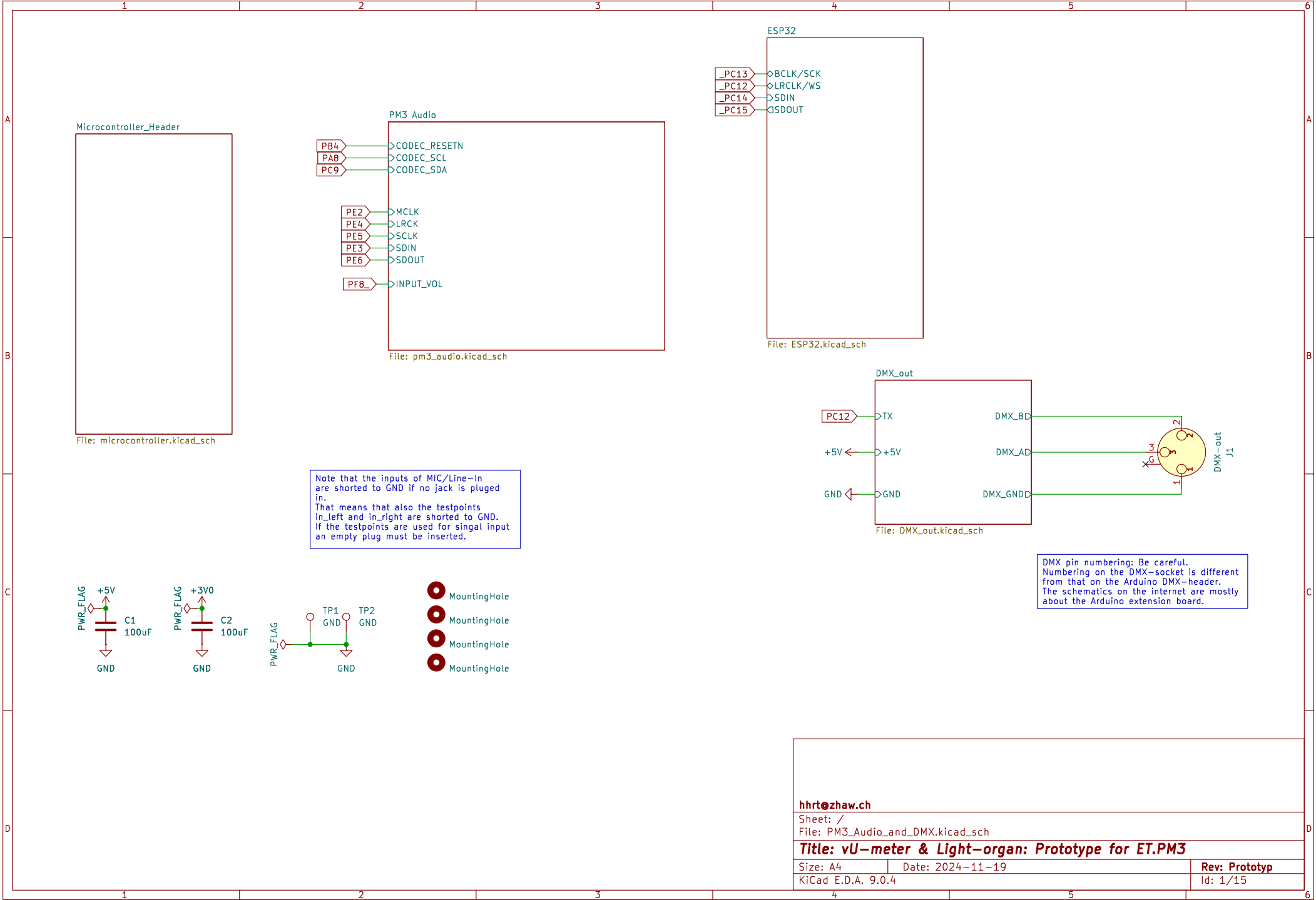
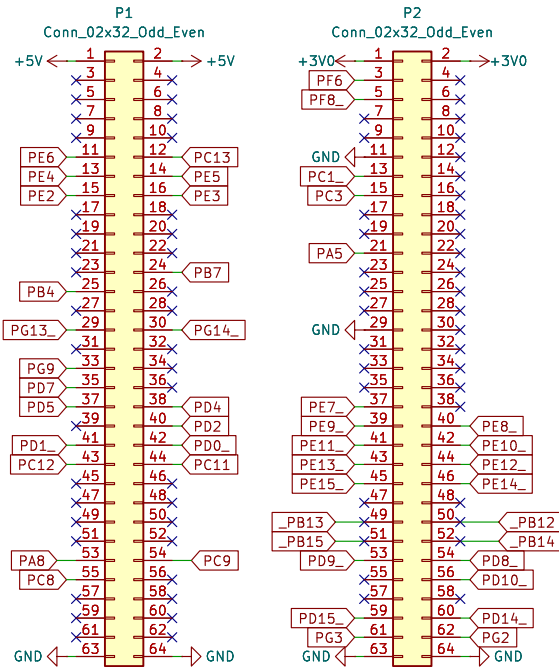






Only a dozen pins from the 128 are in use.
It might be a good idea not to solder the full 2x32 pin sockets
but to use shorter sockets or ones with just one row
(or a combination) to cover the pins in use.
The reason is that otherwise it is quite difficult to disconnect
the microcontroller evalboard from this PCB without breaking it.
What also works well is to plug the socket into the dev-kit and
solder it then.

Microcontroller evaluation board pin headers:
GPIOs with names ending by _ might be used on the board. Check availability

The outline of the extension connectors of the microcontroller evaluation board
with a list of the available functions and peripherals on each pin is in
Microcontroller_STM32F429/Datasheets/Extension_Connectors.xlsx

List of available ADC inputs and DAC output:
PF6 = ADC3_IN4 used for IN_LEFT
PC3 = ADC123_IN13 used for IN_RIGHT
PF8 = ADC3_IN6
PC1 = ADC123_IN11
PA5 = ADC12_IN5 or DAC_OUT2

DMX output to LED party panel:
PC12 = UART5_TX
PC11 = Enable

List of PWM outputs:
PB4 = TIM3_CH1 red
PC8 = TIM3_CH3 green
PE5 = TIM9_CH1 blue
PE6 = TIM9_CH2 white
PB7 = TIM4_CH2 available for special function

USART for other purposes:
PG14 = USART6_TX (with LED RED indicator LD4)
PG9 = USART6_RX

USART for OpenLog (log data on SD card):
PE6 (might be on another pin) = GRN = reset
PA9 = USART1_TX (VCP) = RXI
PA10 = USART1_RX (VCP) = TXO

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Sheet: /Microcontroller_Header/
File: microcontroller.kicad_sch

Title: Microcontroller evaluation board header

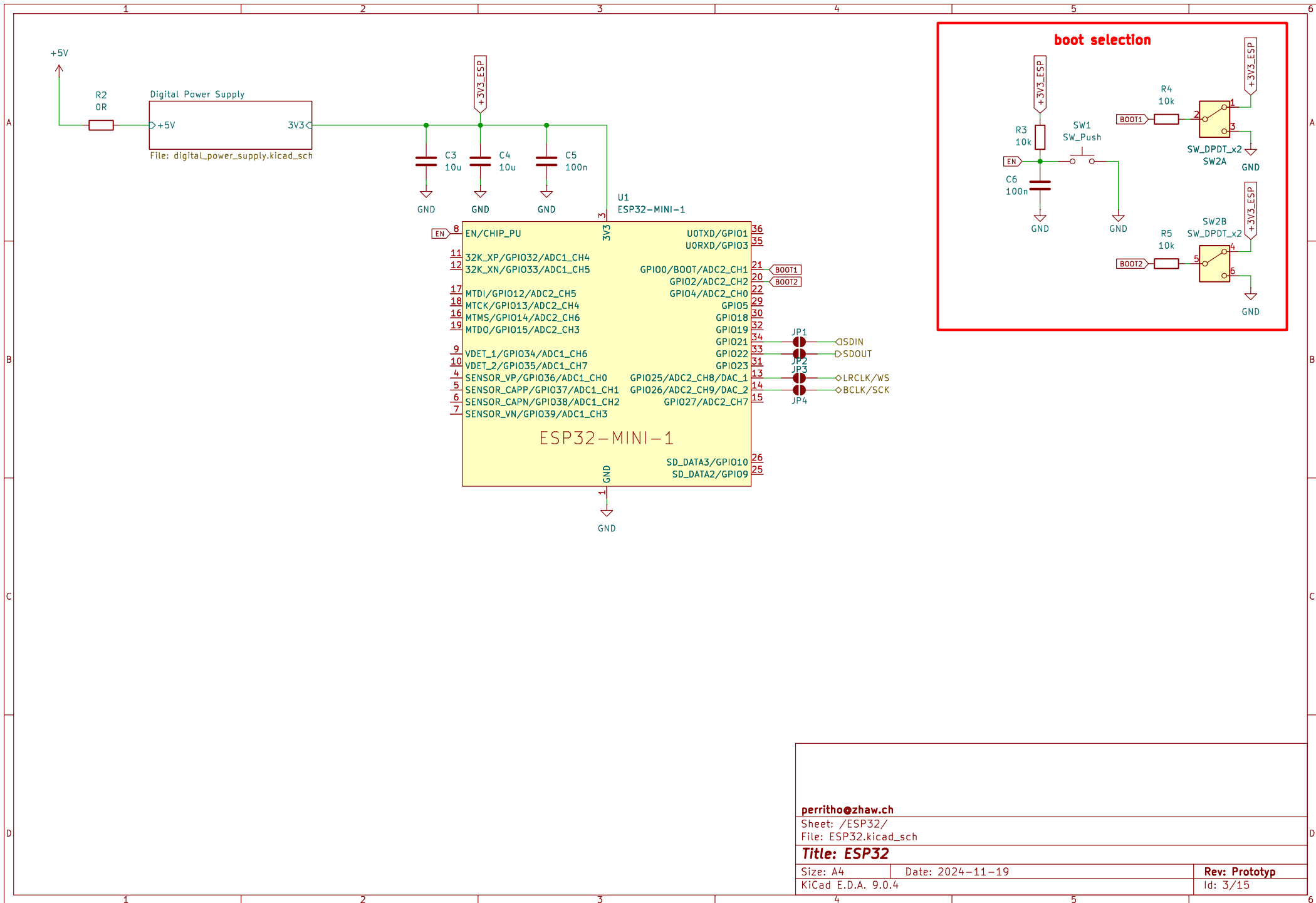
Size: A4

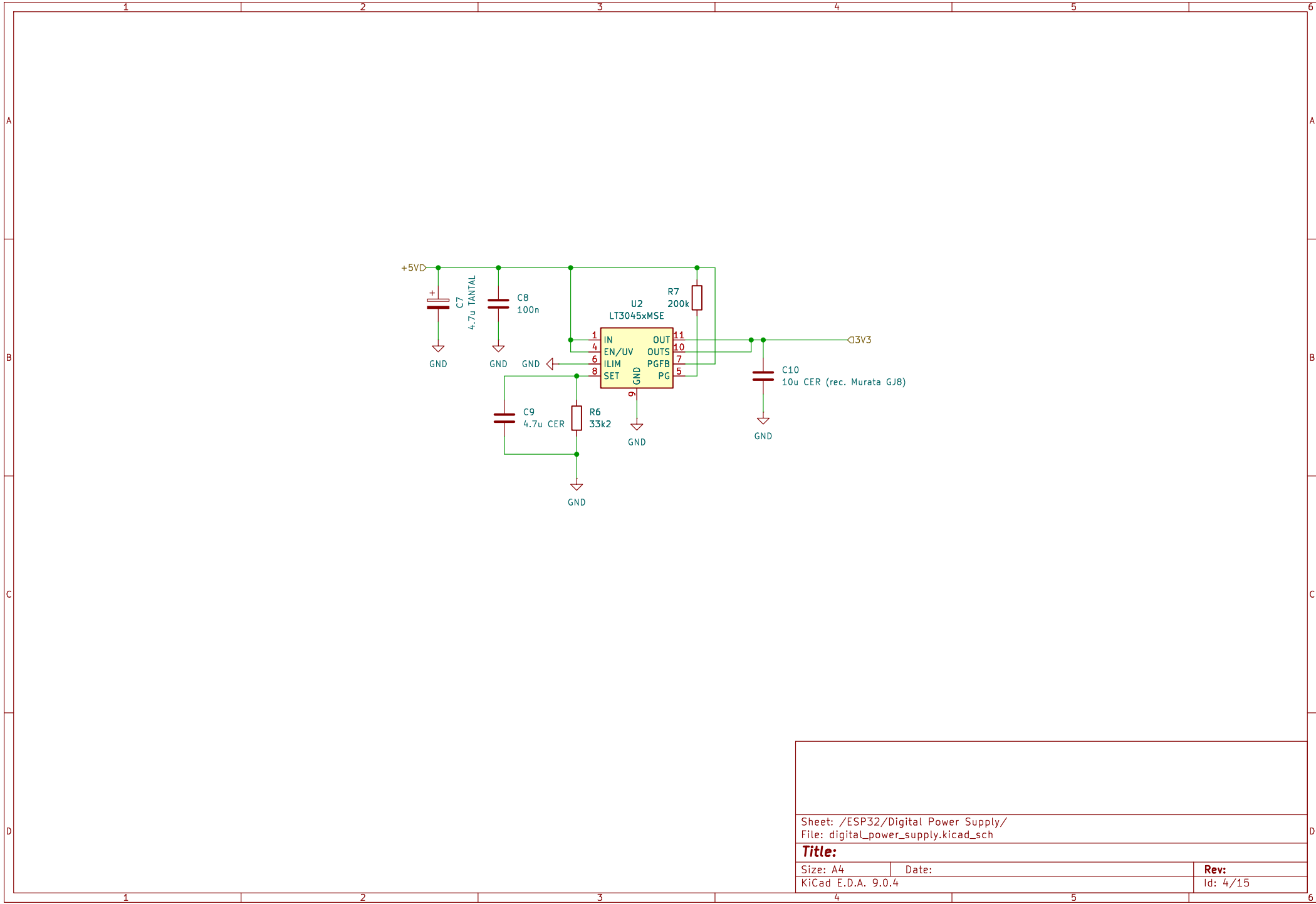
Date: 2024-11-19

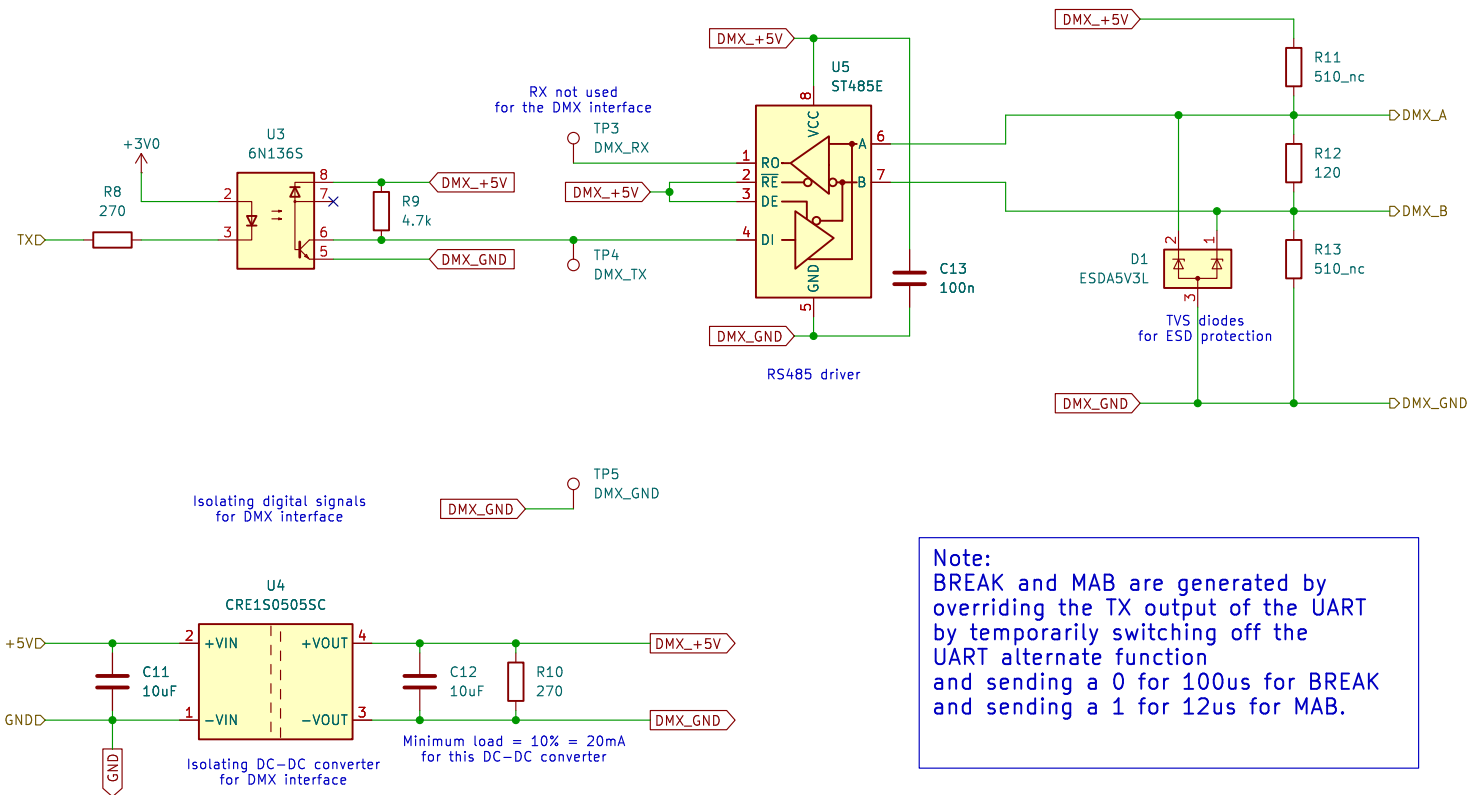
Rev: Prototyp

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Id: 2/15







Note:
BREAK and MAB are generated by overriding the TX output of the UART by temporarily switching off the UART alternate function and sending a 0 for 100us for BREAK and sending a 1 for 12us for MAB.

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Sheet: /DMX_out/
File: DMX_out.kicad_sch

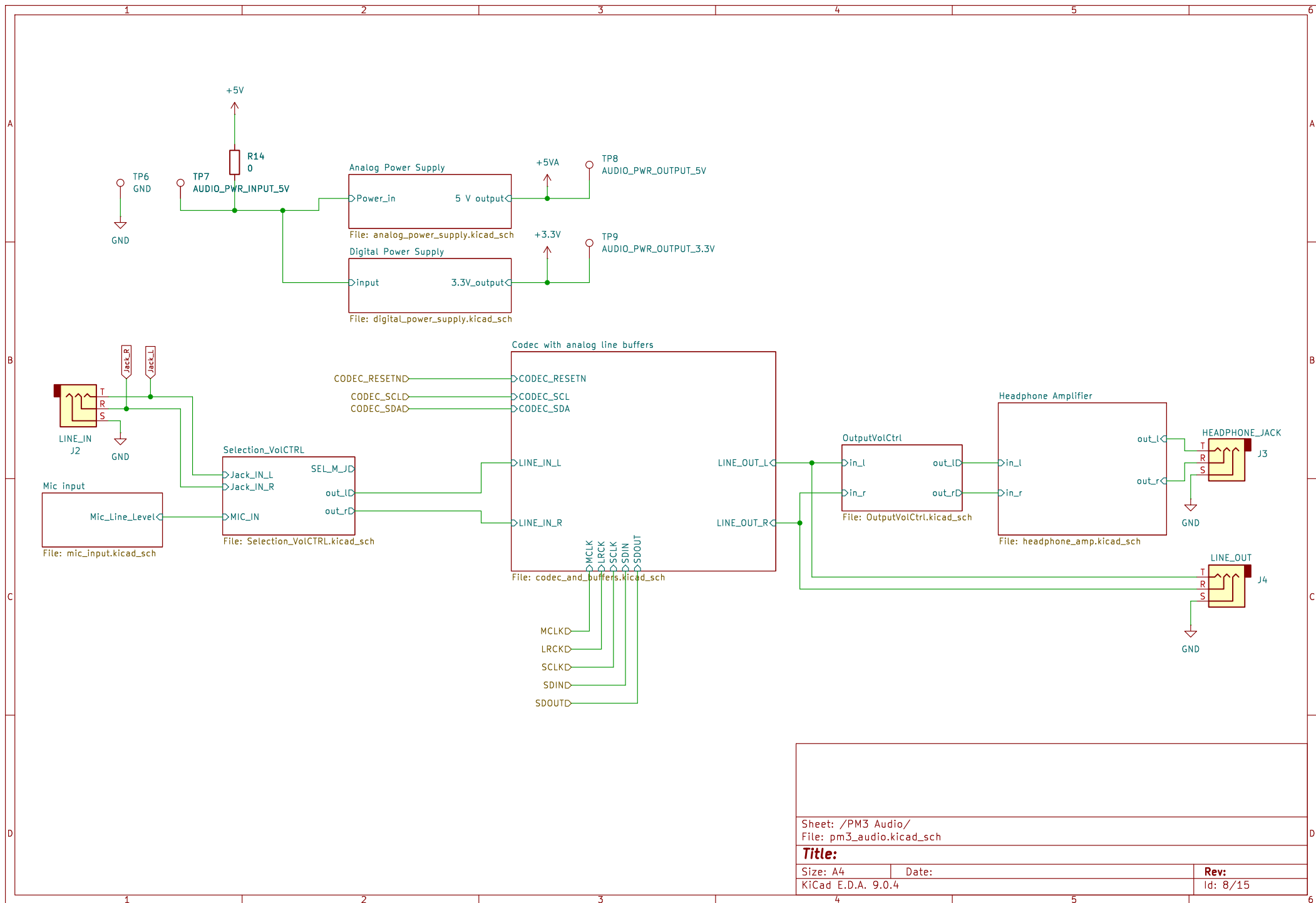
Title: DMX output to LED party panel

Size: A4 Date: 2024-11-19

KiCad E.D.A. 9.0.4

Rev: Prototyp

Id: 5/15



Sheet: /PM3 Audio/
File: pm3_audio.kicad_sch

Title:

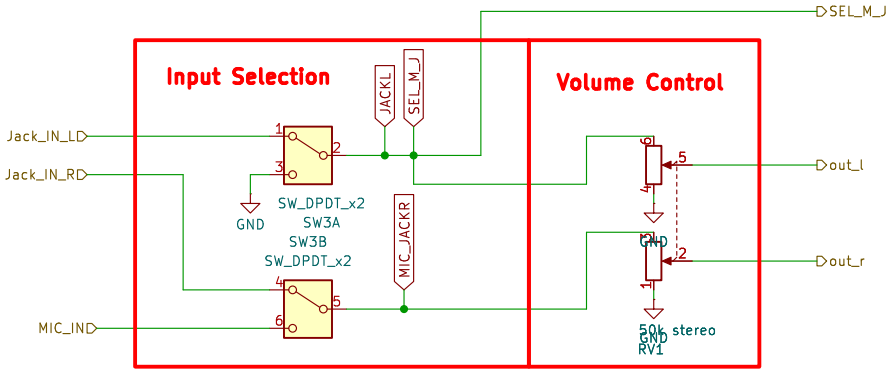
Size: A4
KiCad E.D.A. 9.0.4

Date:

Rev:

Id: 8/15

switches select which audio source is chosen
Firmware knows which one is chosen by reading out the logic level
for pin SEL_M_J if its always 0V then it knows its the microphone.



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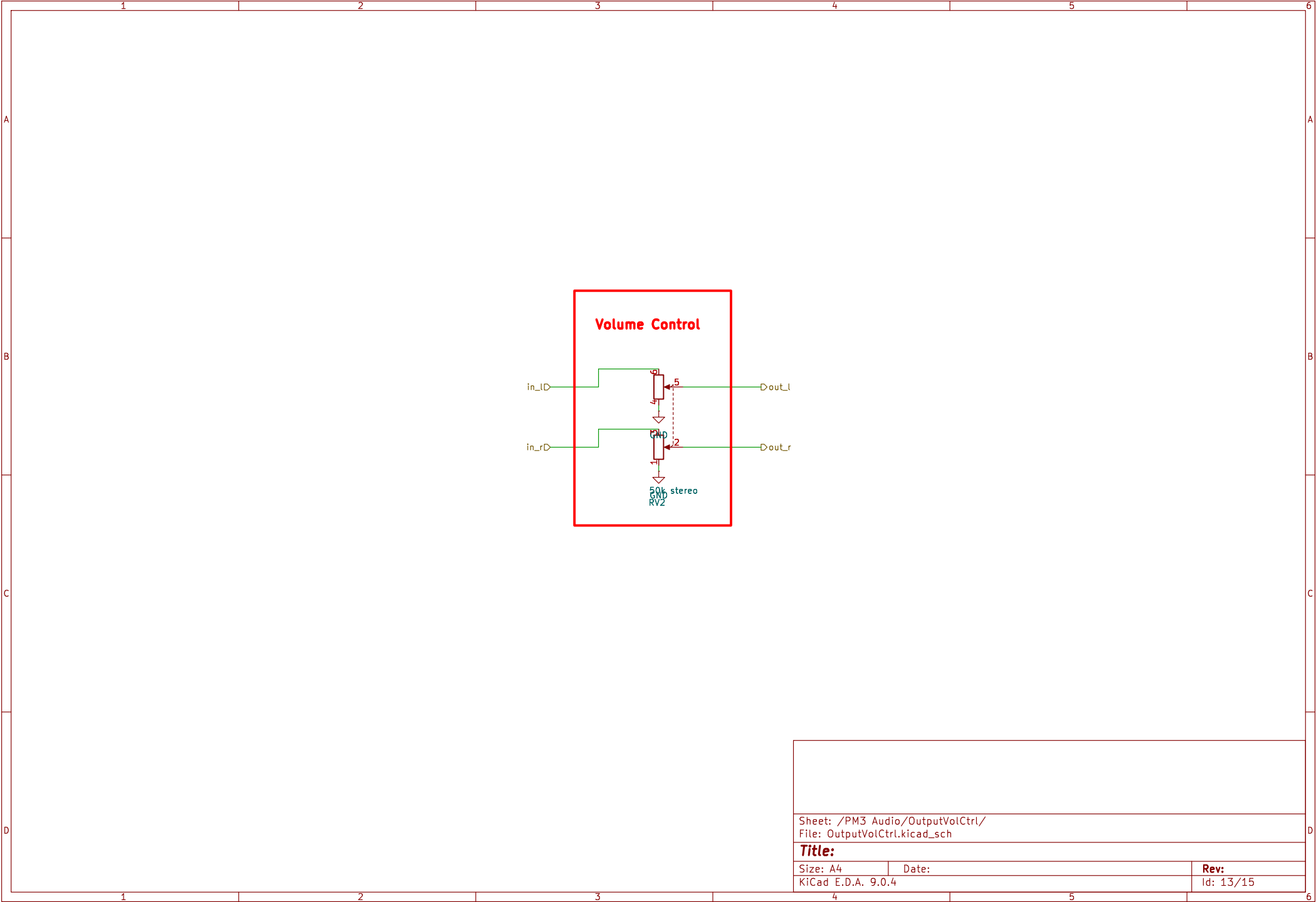
Sheet: /PM3 Audio/Selection_VolCTRL/
File: Selection_VolCTRL.kicad_sch

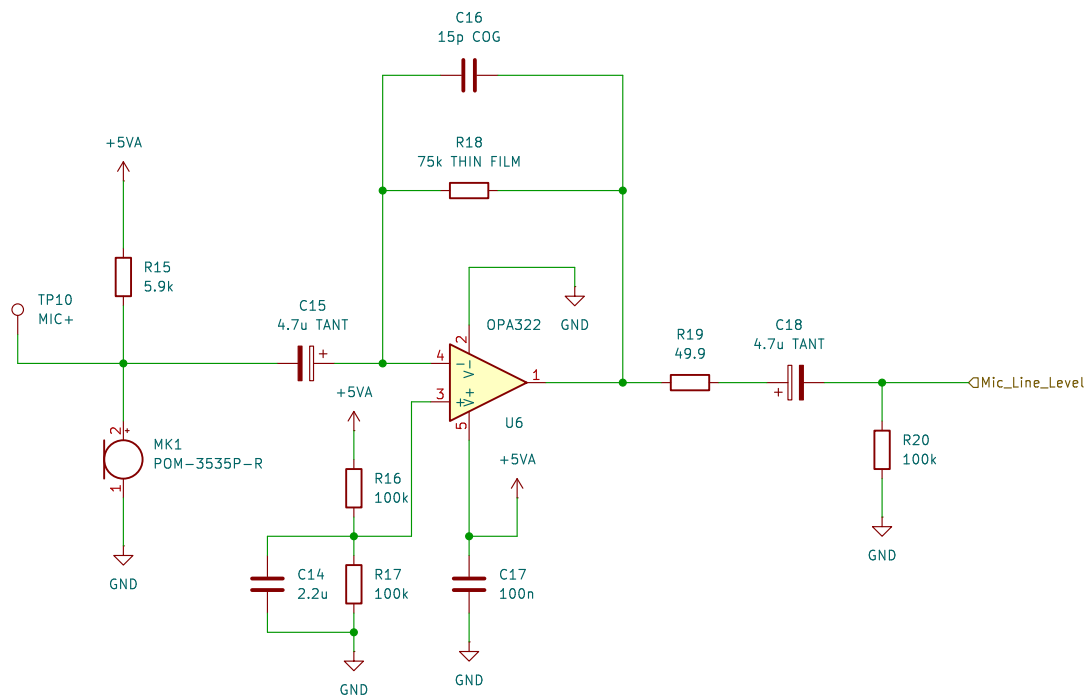
Title: Selection & Volume Control

Size: A4
KiCad E.D.A. 9.0.4

Date: 2025-09-22

Rev: 1
Id: 12/15





Sheet: /PM3 Audio/Mic input/
File: mic_input.kicad_sch

Title:

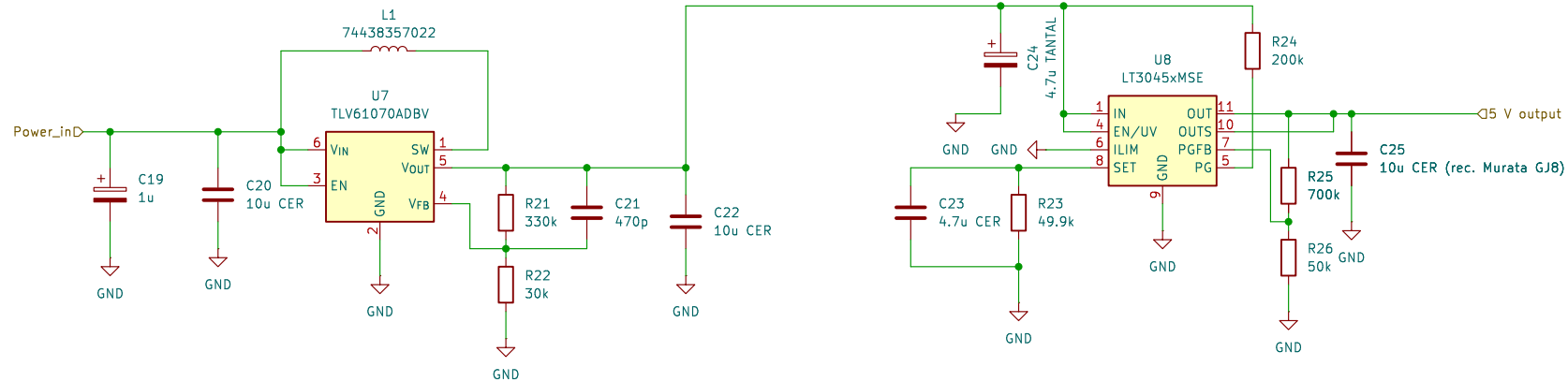
Size: A4

Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 9/15



Sheet: /PM3 Audio/Analog Power Supply/
File: analog_power_supply.kicad_sch

Title:

Size: A4

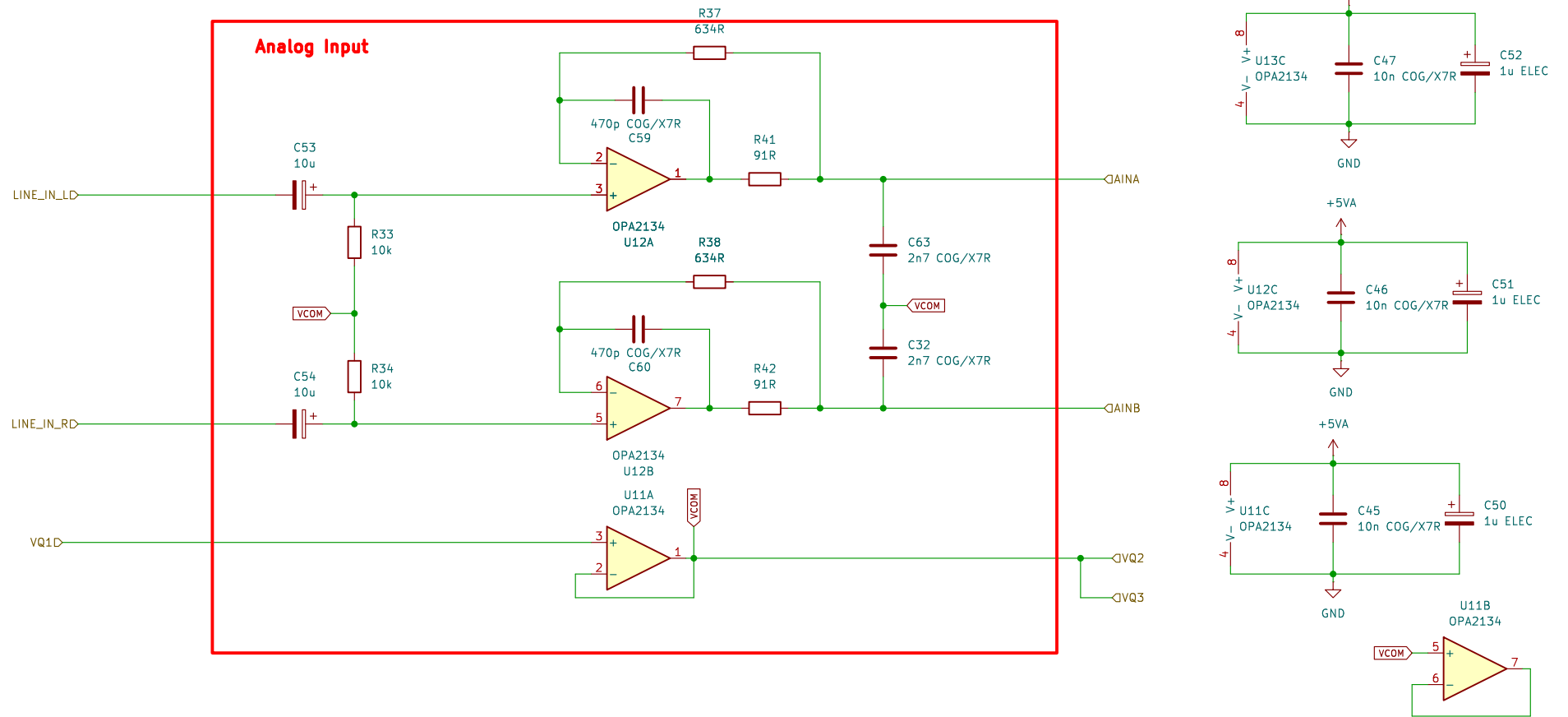
Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 10/15

Id: 11/15



Sheet: /PM3 Audio/Codec with analog line buffers/Audio Input/
File: audio_input.kicad_sch

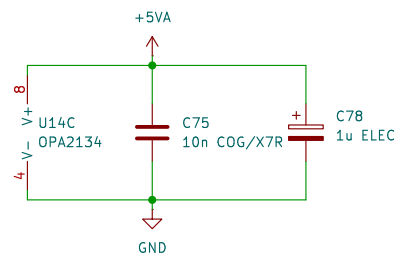
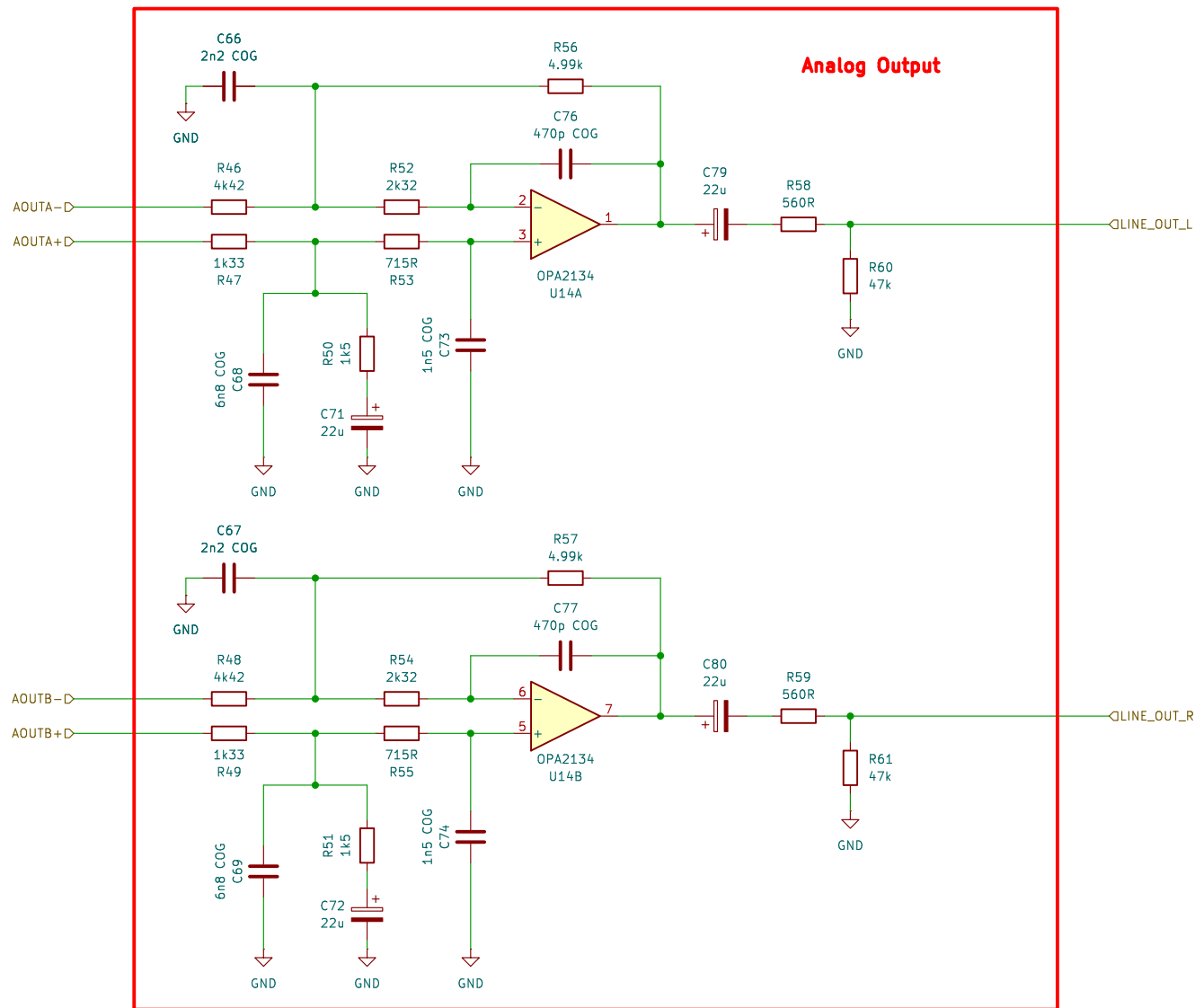
Title:

Size: A4 Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 13/15



Sheet: /PM3 Audio/Codec with analog line buffers/Audio Output/
File: audio_output.kicad_sch

Title:

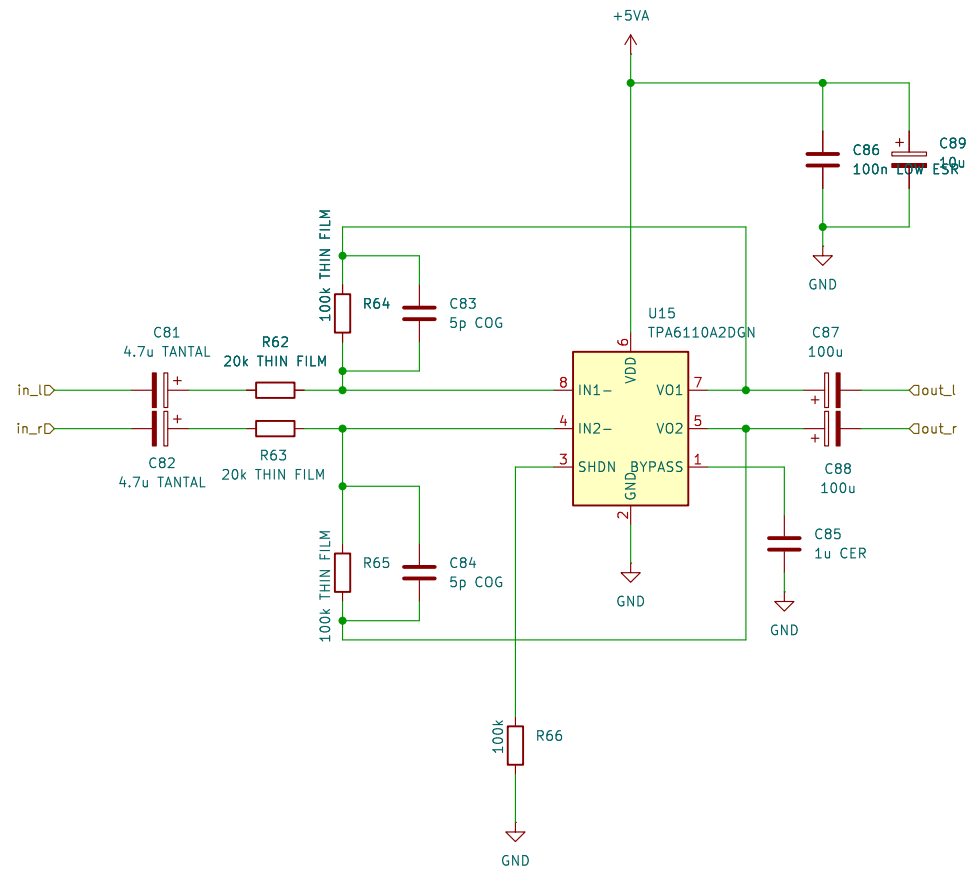
Size: A4

Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 14/15



Sheet: /PM3 Audio/Headphone Amplifier/
File: headphone_amp.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 15/15